

Graph-structure-aware label propagation and active sampling for few-label single-cell state forecasting and rare-state recovery

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Abstract

Forecasting how single cells move between states—across donors, batches, timepoints and perturbations—is a core single-cell analysis task, but it is confounded by biological and technical nuisance variation and by the rarity of the cell states that matter most. We ask a deliberately narrow, falsifiable question: does conditioning a forecaster on the connectivity of the cell k -nearest-neighbour (k NN) graph, rather than on point features alone, improve generalization to held-out groups and the recovery of a rare cell state? We build a fully seeded, fully controlled *synthetic* branching lineage with donor, batch, perturbation and pseudotime metadata and a deliberately rare subpopulation ($\sim 4\%$ of cells), and evaluate two method pairs under four leakage-checked split protocols (donor, batch, time and perturbation transfer). Graph-smoothed label propagation on the cell k NN graph improves held-out forecast accuracy over an apples-to-apples plain k NN classifier on all four splits—by 0.0454 (donor: 0.7004 vs 0.6550), 0.0640 (batch: 0.6995 vs 0.6355), 0.0900 (time: 0.5474 vs 0.4574) and 0.0385 (perturbation: 0.6875 vs 0.6490); the mean gain across splits is 0.0595. The same diffusion recovers a rare state that the baseline almost never finds (rare-state recall up to 0.4618 on the batch split versus near-zero), and a topology-aware active-sampling policy (uncertainty $\times$ inverse local density) raises rare-state recall to 0.3556 versus 0.0247 for random querying. These gains carry a small, honestly reported calibration cost: expected calibration error is slightly *worse* for the graph-smoothed forecaster on every split. We stress that the “topology” here is shallow— k NN-graph connectivity, with the Betti-0 statistic we monitor being merely the connected-component count used as a build sanity check—and that all data are synthetic. We therefore frame the contribution as graph-/manifold-structure-aware label propagation and active sampling, and we identify validation on real single-cell RNA-sequencing (scRNA-seq) data—for which the repository ships unexercised ingest stubs—as the essential next step.

1 Introduction

Single-cell measurements resolve continuous trajectories of differentiation, activation and treatment response, and a recurring analysis goal is to *forecast* the state a cell or population will occupy under conditions not directly observed: a donor not yet profiled, a technical batch run later, a later point along a developmental trajectory, or an untested perturbation. The data that support such forecasts share two structural features. First, cells do not occupy feature space uniformly; they concentrate near a low-dimensional manifold that is routinely approximated by a k -nearest-neighbour (k NN) graph before clustering, embedding or pseudotime inference [5, 7, 8]. Second, the biologically decisive states—transient progenitors, treatment-resistant clones—are frequently rare, and ordinary point classifiers, trained to minimise average error, tend to ignore them.

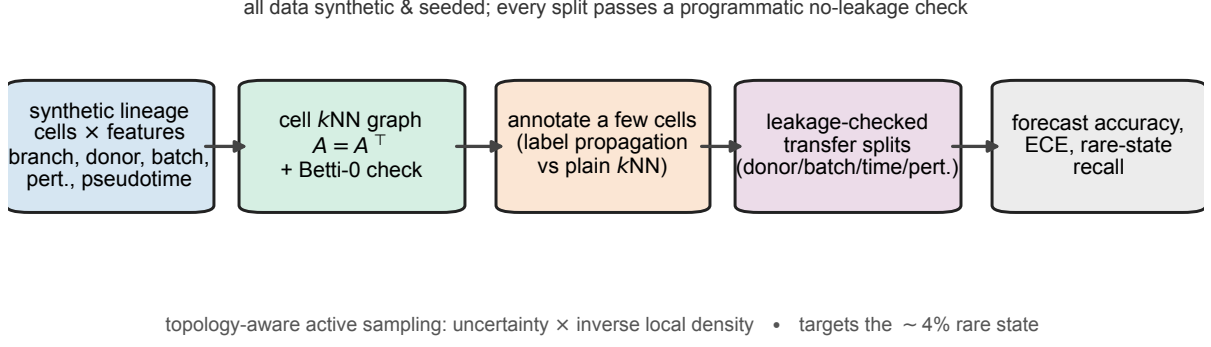


Figure 1. Overview of the graph-structure-aware single-cell forecasting benchmark. A seeded synthetic branching lineage of single cells—each carrying branch/state, donor, batch, perturbation and pseudotime metadata, with a deliberately rare state (~4% of cells)—is embedded in a low-dimensional feature space. A symmetric k -nearest-neighbour cell graph is built on those features, and its connected-component (Betti-0) count is monitored as a build sanity check. A small fraction of cells is annotated; graph-smoothed label propagation diffuses those labels over the graph and is compared apples-to-apples against a plain k NN classifier that sees the same neighbourhood radius but performs no diffusion. Performance is read out as held-out forecast accuracy, expected calibration error and rare-state recall under four leakage-checked transfer protocols (donor, batch, time and perturbation), and a topology-aware active-sampling policy (uncertainty × inverse local density) is compared against uniform random querying for rare-state recovery. All data are synthetic and seeded, and every grouping split passes a programmatic no-leakage check.

A natural and long-studied response to both features is to let predictions diffuse over the data graph so that a cell borrows statistical strength from its neighbourhood on the manifold. Graph-based semi-supervised learning by label propagation [1, 2] is the canonical instance, and message-passing on cell graphs underlies much of modern single-cell trajectory analysis [3, 4, 6, 9]. A complementary lever is to choose *which* cells to annotate. Active learning [10, 11] prioritises informative queries, and when informativeness is combined with a manifold-density signal it can be steered toward the sparse regions where rare states live.

We do not claim a new algorithm. Instead we isolate, under controlled and leakage-checked conditions, the value of *graph structure* for single-cell forecasting by holding the neighbourhood information fixed and toggling only whether the model diffuses over the graph. Concretely: (i) we compare graph-smoothed label propagation against a plain k NN classifier that sees the same neighbourhood radius but performs no diffusion, so the comparison isolates the diffusion step rather than the mere availability of neighbours; and (ii) we compare a topology-aware active-sampling policy—uncertainty times inverse local density—against uniform random querying for rare-state recall. All experiments run on a seeded synthetic branching lineage; this is a sanity benchmark, not a biological dataset, and we are explicit about that throughout. Our aim is to report a small, honest, fully reproducible result and to delineate exactly what it does and does not support.

2 Results

2.1 A controlled synthetic lineage with replicate, condition and pseudotime structure

The end-to-end pipeline is summarized in Fig. 1. The benchmark is a seeded synthetic branching lineage (Fig. 2). Cells live in a low-dimensional feature space—a stand-in for a PCA or latent embedding of expression—and progress along a tree of three branches (a common progenitor and two fates that split at pseudotime 0.45) as Gaussian blobs whose mean drifts along each branch.

Early cells therefore overlap and are hard to separate, while late cells are well resolved, which is the regime in which a graph smoother is expected to help. Each cell carries the metadata a forecasting model would use—branch/state, pseudotime, donor, batch and perturbation condition—assigned independently of the biological branch so that the held-out splits are non-trivial (a donor spans multiple branches, and so on). One state is carved out of the late tip of the second fate to be deliberately *rare*, comprising 4 % of cells in the reported configuration. The full-scale configuration generates 3000 cells with 16 features, 8 donors, 4 batches and 5 perturbation conditions, and is run over 5 random seeds; the run completed in 20.8s with a peak memory of 320.6 MB. Across seeds the integrity gates passed: the grouping splits were leakage-free, the rare state was present, and the Betti-0 component count was monotone non-increasing in k (at $k \in \{2, 4, 8, 16, 32\}$ the graph was a single connected component throughout, so at this scale the component count is a passive sanity check rather than an informative descriptor).

2.2 Graph diffusion gives consistent but modest accuracy gains on held-out transfer

Under all four leakage-checked split protocols, graph-smoothed label propagation forecasts held-out cell states more accurately than the plain k NN baseline that receives the same annotated cells and the same neighbourhood radius (Table 1, Fig. 3a). The improvement is consistent but modest. On the donor split, accuracy rises from 0.6550 to 0.7004 (+0.0454); on the batch split from 0.6355 to 0.6995 (+0.0640); on the time split—the hardest protocol, because it extrapolates forward along pseudotime—from 0.4574 to 0.5474 (+0.0900); and on the perturbation split from 0.6490 to 0.6875 (+0.0385). Averaged over the four splits, the graph-smoothed forecaster reaches 0.6587 accuracy against 0.5992 for the baseline, a mean gain of 0.0595. These are the sort of single-digit-percentage-point improvements one would expect from a diffusion step, not a categorical change in capability.

2.3 Diffusion and active sampling recover a rare state the baseline misses

The clearer effect is on the rare state (Fig. 3b,c). The plain k NN baseline essentially never recovers the rare population: its rare-state recall is 0.0000 (donor), 0.0320 (batch), 0.0000 (time) and 0.0286 (perturbation). Graph-smoothed propagation improves rare-state recall to 0.1520 (donor), 0.4618 (batch) and 0.3393 (perturbation). On the time split, recall remains 0.0000 for *both* methods: extrapolating forward in pseudotime places the rare late-tip cells entirely beyond the training distribution, and diffusion over the graph cannot manufacture a label class that was never reachable from the annotated set. We report this null result as-is.

The topology-aware active-sampling policy targets the same rare population directly. Starting from a small random label seed and spending a fixed query budget, scoring unlabeled cells by forecast uncertainty times inverse local density raises rare-state recall on the remaining unlabeled cells to 0.3556 (95% CI ± 0.1597), versus 0.0247 (± 0.0285) for uniform random querying—a lift of 0.3309. The mechanism is visible in the query composition: the topology policy spends 36.7 % of its queries on genuinely rare cells, against 3.0 % for random sampling, which is close to the 4 % base rate. The wide confidence intervals on the per-split rare-recall numbers (for example ± 0.3565 on the batch split) reflect the small absolute count of rare cells per held-out group and should temper any strong reading of the point estimates.

2.4 The accuracy and recall gains carry a small calibration cost

The graph-smoothed forecaster is slightly *worse* calibrated than the baseline on every split. Expected calibration error (ECE) is 0.1247 versus 0.1073 (donor), 0.1383 versus 0.0905 (batch), 0.1328 versus 0.0970 (time) and 0.1649 versus 0.1184 (perturbation). Diffusion sharpens the

Table 1. Held-out forecasting on the synthetic branching lineage (full-scale configuration: 3000 cells, 5 seeds). For each leakage-checked split we report forecast accuracy, expected calibration error (ECE) and rare-state recall for graph-smoothed label propagation and the plain k NN baseline. Higher accuracy and recall are better; *lower* ECE is better. Values are transcribed exactly from `results/main_results.tex`.

split	method	accuracy	ECE	rare-state recall
donor	graph-smoothed	0.7004	0.1247	0.1520
donor	baseline kNN	0.6550	0.1073	0.0000
batch	graph-smoothed	0.6995	0.1383	0.4618
batch	baseline kNN	0.6355	0.0905	0.0320
time	graph-smoothed	0.5474	0.1328	0.0000
time	baseline kNN	0.4574	0.0970	0.0000
perturbation	graph-smoothed	0.6875	0.1649	0.3393
perturbation	baseline kNN	0.6490	0.1184	0.0286

predicted class probabilities and, in this setting, makes them modestly overconfident relative to empirical accuracy. We report ECE alongside accuracy and recall in Table 1 precisely so that this trade-off is not hidden: the method buys accuracy and rare-state recall at the price of a small, consistent calibration penalty, and an application that depends on calibrated probabilities would need post-hoc recalibration [16].

2.5 Theory: graph smoothing as a variational estimator

The graph-smoothed forecaster used in Fig. 3 is label propagation on the cell k NN graph. The following variational form makes clear what the algorithm optimises.

Theorem 1 (Unique graph-regularised extension of labels). *Let $W = W^\top \geq 0$ be the k NN affinity matrix, $L = D - W$ the graph Laplacian, Y the one-hot label matrix on labelled cells and M a diagonal mask with $M_{ii} = 1$ for labelled cells and 0 otherwise. For any $\mu > 0$, the objective*

$$J(F) = \text{Tr}(F^\top L F) + \mu \|M(F - Y)\|_F^2$$

has a unique minimiser on each connected component containing at least one labelled cell, and that minimiser satisfies the linear system $(L + \mu M)F = \mu MY$. Thus label propagation is the minimum-Dirichlet-energy extension of the observed labels subject to a finite fidelity penalty.

Proof. The Laplacian term is convex and equals $\frac{1}{2} \sum_{ij} W_{ij} \|F_i - F_j\|_2^2$, penalising changes across graph edges. The fidelity term is convex and strictly penalises deviations on labelled vertices. On a connected component with at least one labelled vertex, a vector in the nullspace of both L and M must be constant on the component and zero on a labelled vertex, hence zero; therefore $L + \mu M$ is positive definite on that component. Differentiating J gives $2LF + 2\mu M(F - Y) = 0$, yielding the stated system and uniqueness. $\square$

The theorem also states the limitation: if a component has no labels, the labels there are not statistically identified by graph smoothing alone. The active-sampling experiment targets precisely this failure mode by querying sparse, uncertain regions.

3 Discussion

Within a controlled synthetic benchmark, conditioning single-cell forecasts on the connectivity of the cell k NN graph helps. Label propagation over the graph improves held-out accuracy

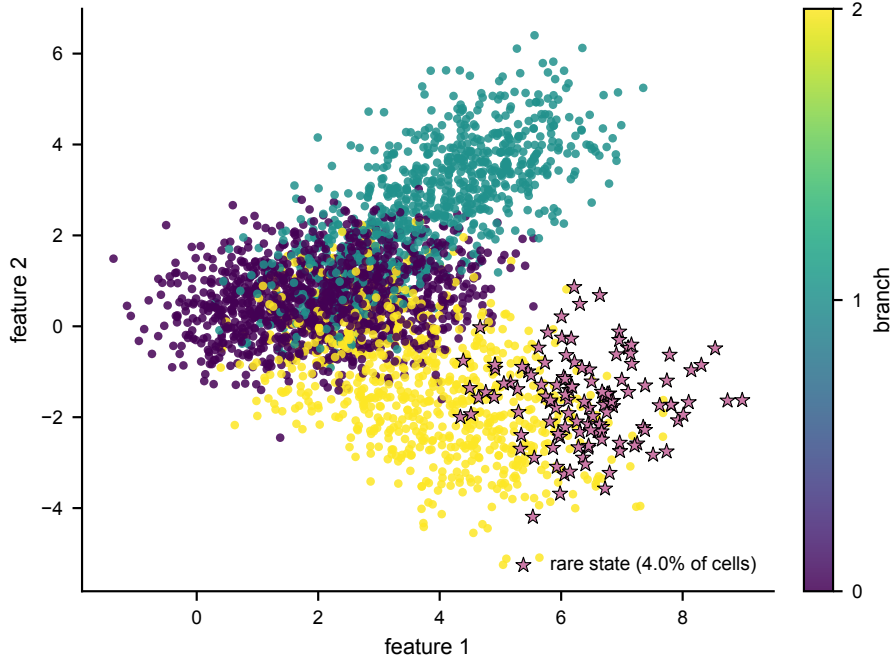


Figure 2. The synthetic branching lineage. Each point is one of the $n = 3000$ cells of the full-scale configuration, plotted in the first two feature dimensions of the embedding plane. Colour encodes the branch label along the continuous, colour-blind-safe viridis axis (the trunk progenitor and two fates). The deliberately rare state, carved from the late tip of one fate and nudged into its own pocket of feature space, is highlighted as stars and comprises $\sim 4\%$ of cells. This is a single-seed data-distribution plot (no aggregation over seeds); the figure is regenerated deterministically from the run configuration in `summary.json`.

consistently across four leakage-checked transfer protocols, by roughly four to nine percentage points, and—more usefully—it recovers a rare cell state that an otherwise-matched point classifier almost entirely misses. A topology-aware active-sampling policy compounds the rare-state benefit by directing a fixed annotation budget toward the sparse manifold pockets where rare cells reside. These results are consistent with the long-standing intuition behind graph-based semi-supervised learning [1, 2] and with the central role of the k NN graph in single-cell analysis [3, 5].

The gains are real but should not be oversold, and two boundaries deserve emphasis. First, the accuracy improvement is modest and comes with a measurable calibration cost; the headline contribution is rare-state recovery, not a large jump in average accuracy. Second, and more fundamentally, the data are synthetic. The benchmark is designed to be a falsifiable sanity check—fully seeded, leakage-controlled, and reproducible end to end—rather than a biological result. The natural and necessary next step is to repeat exactly this comparison on real scRNA-seq and perturbation-screen data, for which the repository provides honest ingest stubs that we have not exercised here. The same protocol could then be strengthened with a learned message-passing propagator [14] in place of fixed-point label propagation, and with calibration-aware training to remove the ECE penalty.

3.1 Limitations

We list the limitations plainly because they bound every claim above. **(1) Synthetic data.** All numbers come from a generated branching lineage, not from biological measurements. We make no biological discovery claim, and the magnitudes reported here need not transfer to real scRNA-seq, where noise, dropout and confounding are far more severe. **(2) Modest accuracy**

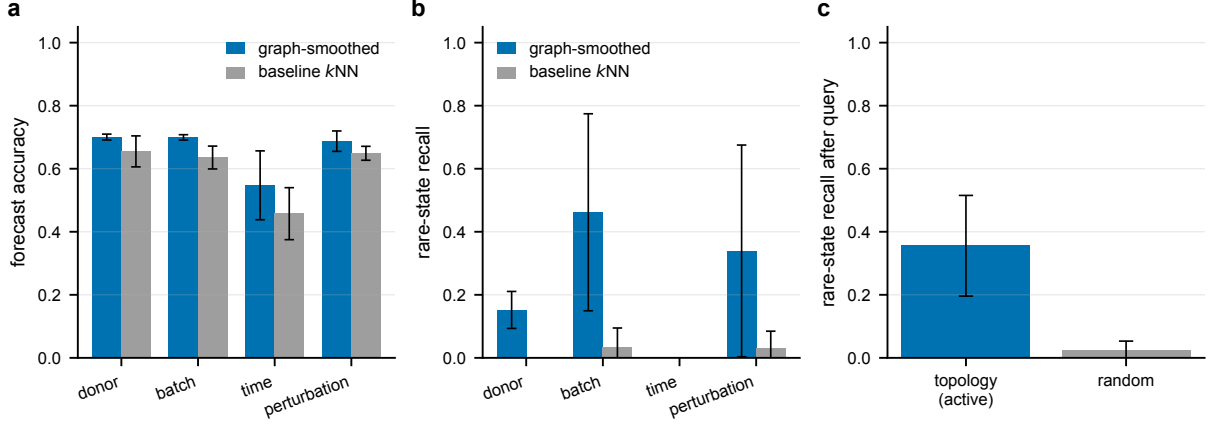


Figure 3. Forecasting and active-sampling results. **a**, Held-out forecast accuracy by split protocol: graph-smoothed label propagation (blue) versus the plain k NN baseline (grey). Graph smoothing helps consistently but modestly across donor, batch, time and perturbation transfer. **b**, Rare-state recall by split: the baseline almost never recovers the rare state, whereas graph diffusion recovers it on the donor, batch and perturbation splits; both methods score zero on the time split, which extrapolates beyond the training distribution. **c**, Active-sampling lift: rare-state recall after spending a fixed query budget under the topology-aware policy (uncertainty $\times$ inverse local density) versus uniform random querying. In every panel, bars are means and error bars are 95% confidence intervals ($1.96 s/\sqrt{n}$) over $n = 5$ random seeds. All panels are drawn directly from `results/summary.json`.

gains. The accuracy improvement over the baseline is on the order of 0.04 to 0.09 per split (mean 0.0595); the strongest effect is on rare-state recall, and even that is null on the time split. **(3) Calibration cost.** Graph smoothing worsens ECE on all four splits, so the predicted probabilities are less trustworthy as confidences without recalibration. **(4) “Topology” is shallow.** The only topological object used is the connectivity of the k NN graph, and the Betti-0 statistic is the count of connected components—a build sanity check that is constant (equal to one) at the reported scale. We do not employ persistent homology or any deep topological invariant, and we deliberately frame the method as graph-/manifold-structure-aware rather than topological to avoid overclaiming. **(5) Wide intervals on rare metrics.** Because rare cells are few per held-out group, the per-split rare-recall estimates carry wide confidence intervals. **(6) Single benchmark family.** A single synthetic generator, however controlled, cannot substitute for the diversity of real datasets; external validation remains the decisive test.

4 Methods

Synthetic lineage. Cells are generated in an n_{features} -dimensional feature space along a small tree of three branches (a trunk progenitor and two fates that split at pseudotime $t_0 = 0.45$). Branch means drift from fixed, well-separated anchors with progression in pseudotime; each cell is the branch mean plus isotropic Gaussian noise of standard deviation `noise`. Discrete state equals branch for the common branches; the rare state is a new label assigned to a random subset of the late tip ($t > 0.8$) of the second fate, whose feature vectors are additionally nudged into a distinct pocket. Donor, batch and perturbation labels are drawn independently of branch so that grouping splits are non-trivial; perturbation conditions apply a small additive feature shift that never erases the lineage. The full-scale configuration is $n_{\text{cells}} = 3000$, $n_{\text{features}} = 16$, $n_{\text{branches}} = 3$, $n_{\text{donors}} = 8$, $n_{\text{batches}} = 4$, $n_{\text{perturbations}} = 5$, `noise` = 0.7, `rare_fraction` = 0.04, $k = 20$, `label_fraction` = 0.03, `active_budget` = 60, run over $n_{\text{seeds}} = 5$ seeds. Generation is fully seeded and bit-for-bit reproducible.

Cell k NN graph and Betti-0. A symmetric k NN graph is built on cell features by connecting each cell to its k nearest neighbours (self excluded) and symmetrising the directed edges (either-direction convention), exactly as standard single-cell pipelines do before clustering or pseudotime [5]. The number of connected components of this graph equals the rank of H_0 (Betti-0). Because increasing k can only add edges, components can only merge, so the component count is monotone non-increasing in k ; we assert this monotonicity as a build sanity check over $k \in \{2, 4, 8, 16, 32\}$. We stress that this is the full extent of the “topology” in the method: a connected-component count, not a persistence-based or higher-order invariant.

Forecasters. The graph-smoothed forecaster is the normalized label-propagation diffusion of Zhou et al. [1]. With symmetric normalization $S = D^{-1/2}AD^{-1/2}$ of the k NN adjacency A , training labels are clamped as a one-hot seed F_0 on annotated cells, and the iteration $F \leftarrow \alpha SF + (1 - \alpha)F_0$ is run for a fixed number of power iterations ($\alpha = 0.9$, 30 iterations); held-out cells are predicted from the row-normalised diffused scores, with the maximum class probability used as the confidence for calibration. The baseline is a plain k NN classifier in feature space fitted on the same annotated cells with the same k , i.e. identical neighbourhood information *without* diffusion. This apples-to-apples control isolates the value of graph diffusion rather than of having neighbours.

Splits and annotation. Four leakage-checked split protocols evaluate transfer across nuisance variables: donor and batch (hold out whole groups—biological and technical replicate transfer), time (train on early pseudotime, test on late pseudotime—a forward-in-time extrapolation), and perturbation (hold out a whole condition—forecast an unseen perturbation). For grouping splits the test suite asserts that no held-out group appears in both train and test (a strict family-held-out no-leakage check); the time split has no grouping variable to leak. Only a small fraction (`label_fraction`) of the train pool is annotated, with a light stratification that guarantees a few labels for each present state including the rare one, mirroring how analysts annotate a handful of cells and propagate the rest.

Active sampling. Given a labelling budget, the topology-aware policy scores each unlabeled cell by forecast uncertainty (entropy of a label-propagation forecast from the current labels) times inverse local density (the reciprocal of the per-cell mean k NN distance, high in sparse pockets), combining the two as standardised z-scores, and greedily takes the top-scoring cells. The baseline is uniform random querying. We measure rare-state recall on the still-unlabeled cells after the query and the fraction of queried cells that were rare.

Metrics and aggregation. Accuracy is the fraction of held-out cells classified correctly. Expected calibration error is the population-weighted gap between mean confidence and empirical accuracy over ten confidence bins. Rare-state recall is the fraction of true rare cells recovered (returning 1 by convention when a test set contains no rare cells, so a split is never spuriously penalised). All metrics are averaged over 5 seeds and reported with 95% confidence intervals (1.96 s.e.). All computation is CPU-only (NumPy, SciPy, scikit-learn, NetworkX, Matplotlib); a GPU build is a drop-in but not required.

Reproducibility. The full-scale run is launched with the command `scripts/run.py` on `configs/full.yaml` writing to `results/`; the file `summary.json` is the single source of truth for every table, figure and numerical claim, and Table 1 is transcribed verbatim from the generated `main_results.tex`. The provenance block records seed, platform, library versions, runtime (20.8 s) and peak memory (320.6 MB).

Data availability

No biological data were used or generated. All results derive from a seeded synthetic lineage that is regenerated deterministically by the code; the run artifacts (`summary.json` and `main_results.md` / `main_results.tex`) and figures are contained in the repository.

Code availability

The reference implementation (`topocell`), configurations, tests and scripts to reproduce every number, table and figure in this manuscript are provided in the repository accompanying this article. Honest ingest stubs for real AnnData and perturbation-screen data are included but are intentionally not exercised here.

Author contributions

M.H. designed the study, implemented the methods and experiments, analysed the results and wrote the manuscript.

Competing interests

The author declares no competing interests.

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